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Internet Of Things: Homework Exercise 3

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1 Assumptions

A RFID system based on Dynamic Frame ALOHA is composed of $N=4$ tags

2 Question 1

Find the overall collision resolution efficiency η in the different cases in which the initial frame size is set to $r_1=1,2,3,4,5,6$

Assume that after the first frame, the frame size is correctly set to the current backlog size

Assume as given the duration of the arbitration period with $N=2,3$ tags when $r=N$ ($L_2=4$, $L_3=51/8$)

We already have this data

$$L_2 = 4$$

$$L_3 = \frac{51}{8} = 6.375$$

To calculate $E[L_{r_1}]$ we need L_4 so let's calculate Let's calculate

$$P(S=0) = \left(\frac{1}{4}\right)^4 \cdot 4 + \left(\frac{1}{4}\right)^4 \cdot \binom{4}{2} \cdot \frac{4!}{2 \cdot 2} = \frac{10}{64}$$

$$P(S=1) = \left(\frac{1}{4}\right)^4 \cdot 4 \cdot 4 \cdot 3 = \frac{3}{16}$$

$$P(S=2) = \left(\frac{1}{4}\right)^4 \cdot \binom{4}{2} \cdot 4! = \frac{9}{16}$$

and finally

$$L_4 = 4 + P(S=4) \cdot L_4 + P(S=3) \cdot L_3 + P(S=2) \cdot L_2$$

$$= 4 + \frac{10}{64} \cdot L_4 + \frac{3}{16} \cdot L_3 + \frac{9}{16} \cdot L_2$$

Solving for L_4 we get

$$L_4 - \frac{5}{32}L_4 = 4 + \frac{3}{16} \cdot \frac{51}{8} + \frac{9}{16} \cdot 4$$

$$L_4 = \frac{953}{108} \simeq 8.824$$

Now for case r_1 we have

$$E[L] = 1 + 1 \cdot L_4 \simeq 9,824$$

$$\eta_1 = \frac{4}{9.824} \simeq 0.407$$

For case r_2 we have

$$\begin{aligned}
 P(S=0) &= \left(\frac{1}{2}\right)^4 \cdot 2 + \left(\frac{1}{2}\right)^4 \cdot \binom{4}{2} = \frac{1}{2} \\
 P(S=1) &= \left(\frac{1}{2}\right)^4 \cdot 4 \cdot 2 = \frac{1}{2} \\
 E[L] &= 2 + \frac{1}{2}L_4 + \frac{1}{2}L_3 = 2 + 4.412 + 3.187 \simeq 9.599 \\
 \eta_2 &= \frac{4}{9.599} \simeq 0.417
 \end{aligned}$$

For case r_3 we have

$$\begin{aligned}
 P(S=0) &= \left(\frac{1}{3}\right)^4 \cdot [3 + \binom{3}{2} \frac{4!}{2 \cdot 2}] = \frac{7}{27} \\
 P(S=1) &= \left(\frac{1}{3}\right)^4 \cdot (4 \cdot 3 \cdot 2) = \frac{8}{27} \\
 P(S=2) &= \left(\frac{1}{3}\right)^4 \cdot [\binom{3}{2} (4 \cdot 3)] = \frac{12}{27} \\
 E[L] &= 3 + \frac{7}{27}L_4 + \frac{8}{27}L_3 + \frac{12}{27}L_2 \simeq 8.954 \\
 \eta_3 &= \frac{4}{8.954} \simeq 0.447
 \end{aligned}$$

For case r_4 we have

$$\begin{aligned}
 E[L] &= 8.824 \\
 \eta_4 &= \frac{4}{8.824} \simeq 0.453
 \end{aligned}$$

For case r_5 we have

$$\begin{aligned}
 P(S=0) &= \left(\frac{1}{5}\right)^4 \cdot [5 + \binom{5}{2} \frac{4!}{2 \cdot 2}] = \frac{13}{125} \\
 P(S=1) &= \left(\frac{1}{5}\right)^4 \cdot (4 \cdot 5 \cdot 4) = \frac{16}{125} \\
 P(S=2) &= \left(\frac{1}{5}\right)^4 \cdot [\binom{5}{2} (4 \cdot 3) \cdot 3] = \frac{72}{125} \\
 E[L] &= 5 + \frac{13}{125}L_4 + \frac{16}{125}L_3 + \frac{72}{125}L_2 \simeq 9.038 \\
 \eta_5 &= \frac{4}{9.038} \simeq 0.443
 \end{aligned}$$

For case r_6 we have

$$P(S = 0) = \left(\frac{1}{6}\right)^4 \cdot \left[6 + \binom{6}{2} \frac{4!}{2 \cdot 2}\right] = \frac{2}{27}$$

$$P(S = 1) = \left(\frac{1}{6}\right)^4 \cdot (4 \cdot 6 \cdot 5) = \frac{5}{54}$$

$$P(S = 2) = \left(\frac{1}{6}\right)^4 \cdot \left[\binom{6}{2} (4 \cdot 3) \cdot 4\right] = \frac{5}{9}$$

$$E[L] = 6 + \frac{2}{27}L_4 + \frac{5}{54}L_3 + \frac{5}{9}L_2 \simeq 9.466$$

$$\eta_6 = \frac{4}{9.466} \simeq 0.423$$

3 Question 2

After computing the values of the efficiency with the different frame sizes, produce a plot with values of η over r_1 (as figure) -> add in the report

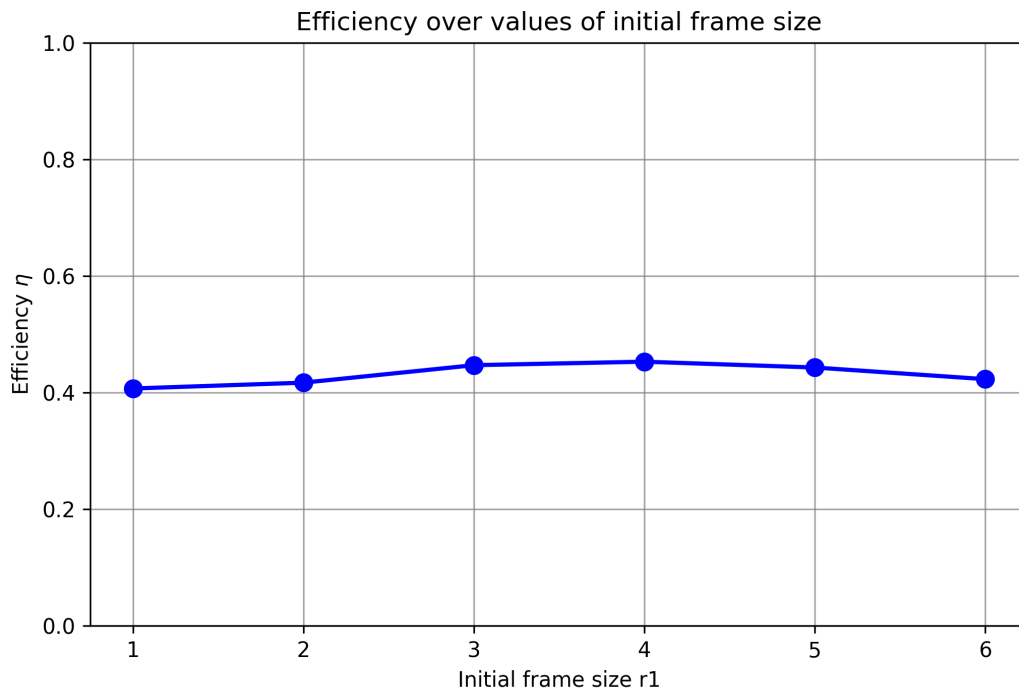


Figure 1: Graph

4 Question 3

For what values of r_1 we have the maximum value for η ? Briefly comment this.

Based on the generated plot and the calculations performed, the maximum value for the efficiency η is obtained for $r_1 = 4$ when $\eta \simeq 0.453$. This confirms the fundamental theoretical principle of Dynamic Frame ALOHA that the efficiency is maximized when the frame size (r) is equal to the number of tags (N).

If we were to choose an $r < 4$, we would increase the probability of collisions among the tags.

If we were to choose an $r > 4$, we would introduce an excessive number of empty slots and therefore waste channel resources.